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INFORMATION PROCESSING DEVICE, CONTROL DEVICE, AND CONTROL METHOD

Abstract

An information processing device includes a first Josephson parametric oscillator used as an information processing qubit element, a second Josephson parametric oscillator used as a readout qubit element, each configured with a Josephson parametric oscillator, and a coupler coupling the information processing qubit element and the readout qubit element, wherein a bit value indicated by the readout qubit element is same as a bit value indicated by the information processing qubit element.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is based upon and claims the benefit of priority from Japanese patent application No. 2024-029419, filed on Feb. 29, 2024, the disclosure of which is incorporated herein in its entirety by reference.

TECHNICAL FIELD

[0002] The present disclosure relates to an information processing device, a control device, and a control method.

BACKGROUND ART

[0003] Information processing devices that use qubit devices, such as quantum annealing machines, have been proposed.

[0004] For example, Japanese Unexamined Patent Application, Publication No. 2021-132188 (Patent Document 1) discloses an information processing device that uses a nonlinear oscillator such as a Kerr parametric oscillator.

SUMMARY

[0005] It is desirable to read out the bit value from the qubit device with the highest possible accuracy. In a case where the degree of freedom of control over a qubit device targeted for reading out bit values is high, it is expected that the control can be optimized for reading out bit values of the qubit device, allowing the bit values to be read out with relatively high accuracy.

[0006] An example of an object of the present disclosure is to provide an information processing device, a control device, an information processing system, and a control method capable of solving the problems mentioned above.

[0007] According to a first example aspect of the present disclosure, an information processing device includes an information processing qubit device and a readout qubit device, each configured with a Josephson parametric oscillator, and a coupler that couples the information processing qubit device and the readout qubit device. The readout qubit device is coupled by the coupler to the information processing qubit device so as to assume the same bit value as the information processing qubit device.

[0008] According to a second example aspect of the present disclosure, a control device includes a control means that controls an information processing device including an information processing qubit device and a readout qubit device, each configured with a Josephson parametric oscillator, and a coupler that couples the information processing qubit device and the readout qubit device, so that the readout qubit device is coupled by the coupler to the information processing qubit device so as to assume the same bit value as the information processing qubit device.

[0009] According to a third example aspect of the present disclosure, an information processing system includes an information processing device, and a control device. The information processing device includes an information processing qubit device and a readout qubit device, each configured with a Josephson parametric oscillator, and a coupler that couples the information processing qubit device and the readout qubit device. The control device includes a control means that controls the information processing device so that the readout qubit device is coupled by the coupler to the information processing qubit device so as to assume the same bit value as the information processing qubit device.

[0010] According to a fourth example aspect of the present disclosure, a control method includes a step, performed by a control device that controls an information processing device including an information processing qubit device and a readout qubit device, each configured with a Josephson parametric oscillator, and a coupler that couples the information processing qubit device and the readout qubit device, of controlling the information processing device so that the readout qubit

device is coupled by the coupler to the information processing qubit device so as to assume the same bit value as the information processing qubit device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. **1** is a diagram showing a configuration example of an information processing system according to at least one of example embodiments.

[0012] FIG. **2** is a diagram showing a first example of the state of the output signal from a qubit device according to at least one of the example embodiments.

[0013] FIG. **3** is a diagram showing a second example of the state of the output signal from the qubit device according to at least one of the example embodiments.

[0014] FIG. **4** is a diagram showing a configuration example of a control device according to at least one of the example embodiments.

[0015] FIG. **5** is a diagram showing a first example of a combination of a computational qubit device and a readout qubit device according to at least one of the example embodiments.

[0016] FIG. **6** is a diagram showing a second example of a combination of the computational qubit device and the readout qubit device according to at least one of the example embodiments.

[0017] FIG. **7** is a diagram showing a third example of a combination of the computational qubit device and the readout qubit device according to at least one of the example embodiments.

[0018] FIG. **8** is a diagram showing an example of control timing for the computational qubit device and the readout qubit device according to at least one of the example embodiments.

[0019] FIG. **9** is a diagram showing an example of a processing procedure performed by the control device according to at least one of the example embodiments.

[0020] FIG. **10** is a diagram showing a first example of a configuration of an information processing device where an LHZ model is used, in at least one of the example embodiments.

[0021] FIG. **11** is a diagram showing a second example of a configuration of the information processing device where an LHZ model is used, in at least one of the example embodiments.

[0022] FIG. **12** is a diagram showing a configuration example of the information processing device according to at least one of the example embodiments.

[0023] FIG. **13** is a diagram showing an example of a configuration of the information processing device according to at least one of the example embodiments, where information processing qubit devices and readout qubit devices are denoted as qubit devices, explicitly exemplifying that the information processing device includes multiple qubit devices.

[0024] FIG. **14** is a diagram showing a configuration example of a control device according to at least one of the example embodiments.

[0025] FIG. **15** is a diagram showing a configuration example of an information processing system according to at least one of example embodiments.

[0026] FIG. **16** is a diagram showing a configuration of the information processing system according to at least one of the example embodiments, where information processing qubit devices and readout qubit devices are denoted as qubit devices, explicitly exemplifying that the information processing device includes multiple qubit devices.

[0027] FIG. **17** is a diagram showing an example of a processing procedure in a control method according to at least one of the example embodiments.

[0028] FIG. **18** is a diagram showing a configuration example of a computer according to at least one of the example embodiments.

EXAMPLE EMBODIMENT

[0029] Hereinafter, example embodiments of the present disclosure will be described, however, the present disclosure within the scope of the claims is not limited by the following example

embodiments. Furthermore, not all the combinations of features described in the example embodiments are essential for the solving means of the disclosure.

First Example Embodiment

[0030] FIG. 1 is a diagram showing a configuration example of an information processing system according to at least one of example embodiments. In the configuration shown in FIG. 1, an information processing system 1 includes an information processing device 100 and a control device 200. The information processing device 100 includes multiple qubit devices 110 and a coupler 120. The control device 200 includes a control unit 210 and an observation unit 220.

[0031] The information processing device 100 uses the qubit devices 110 to perform computations.

[0032] The qubit device referred to here is hardware used to represent the value of a qubit (bit value). A Josephson Parametric Oscillator (JPO) may be employed as the qubit device 110.

[0033] The following describes an example in which the information processing system 1 performs quantum annealing. However, the computations performed by the information processing system 1 are not limited to those carried out using a specific method. For example, the information processing system 1 may be configured to function as a quantum gate computer.

[0034] The multiple qubit devices 110 included in the information processing device 100 include a computational qubit device 111 and a readout qubit device 112.

[0035] The computational qubit device 111 is a qubit device 110 that is used to perform computations by quantum annealing.

[0036] The readout qubit device 112 is provided to read out the bit value indicated by the computational qubit device 111. The readout qubit device 112 is not directly used in performing computations using quantum annealing. One readout qubit device 112 is controlled to indicate the same bit value as the bit value indicated by one computational qubit device 111.

[0037] The bit value indicated by a qubit device is also referred to as the bit value of the qubit device.

[0038] The condition where the readout qubit device 112 indicates the same bit value as the computational qubit device 111 is also referred to as the readout qubit device 112 indicating the same bit value as the computational qubit device 111.

[0039] Here, it is possible that control for quantum annealing may result in the computational qubit device 111 not being in a state appropriate for reading out the bit value. On the other hand, control distinct from that used for quantum annealing can be applied to the readout qubit device 112. In this regard, it is expected that the readout qubit device 112 can be controlled to achieve a state appropriate for reading out the bit value.

[0040] FIG. 2 is a diagram showing a first example of the state of the output signal from the qubit device 110. FIG. 2 is a diagram showing the output signal from the qubit device 110 in a case where the state of the qubit device 110 is not in a state appropriate for reading out the bit value, separated into an in-phase amplitude component and a quadrature-phase amplitude component.

[0041] The photon referred to here is a microwave photon that oscillates within the qubit device 110 and is also output from the qubit device 110.

[0042] The horizontal axis of the graph of FIG. 2 represents the in-phase amplitude (I amplitude), and the vertical axis represents the quadrature-phase amplitude (Q amplitude). In the graph of FIG. 2, the output signal from the qubit device 110 is measured multiple times, and points are plotted on the coordinate values of the in-phase amplitude (I amplitude) and quadrature-phase amplitude (Q amplitude) for each measurement.

[0043] In the graph of FIG. 2, the reference signal is set to a phase that is orthogonal to the oscillation output of the qubit device 110. The Q amplitude is shown as a positive value in the case where one of the two coherent states (referred to as first coherent state) is read. Also, the Q amplitude is shown as a negative value in the case where the second coherent state, which has a phase difference of π from the first coherent state, is read from the two coherent states that the qubit device 110 can take.

[0044] Moreover, in the graph of FIG. 2, the distance from the origin (radial distance) corresponds to the oscillation amplitude, and the deflection angle corresponds to the phase of the signal.

[0045] Additionally, the graph of FIG. 2 shows the results of numerical computations, with separate plots for the case where the correct answer is the first coherent state (white dots) and the case where the correct answer is the second coherent state (black dots).

[0046] In the example of FIG. 2, in a case where the photon count is small, the signal amplitude becomes smaller, and the Q amplitude decreases. Additionally, in a case where the noise is large, the uncertainty between the I amplitude and the Q amplitude increases. In the case where the noise is large relative to the signal amplitude, the distributions of the first coherent state and the second coherent state overlap, making it impossible to identify which coherent state it is in based on the I and Q amplitudes obtained from the measurement in the overlapping region.

[0047] FIG. 3 is a diagram showing a second example of the state of the output signal from the qubit device **110**. FIG. 3 is a diagram showing the output signal from the qubit device **110** in a case where the state of the qubit device **110** is in a state more appropriate for reading out the bit value, separated into an in phase amplitude component and a quadrature-phase amplitude component.

[0048] The horizontal axis of the graph of FIG. 3 represents the in-phase amplitude (I amplitude), and the vertical axis represents the quadrature-phase amplitude (Q amplitude). In the graph of FIG. 3, the output signal from the qubit device **110** is measured multiple times, and points are plotted on the coordinate values of the in-phase amplitude (I amplitude) and quadrature-phase amplitude (Q amplitude) for each measurement.

[0049] In the graph of FIG. 3, the reference signal is set to a phase that is orthogonal to the oscillation output of the qubit device **110**. The Q amplitude is shown as a positive value in the case where the first coherent state is read, and the Q amplitude is shown as a negative value in the case where the qubit device **110** is read as the second coherent state.

[0050] Moreover, in the graph of FIG. 3, the distance from the origin (radial distance) corresponds to the oscillation amplitude, and the deflection angle corresponds to the phase of the signal.

[0051] Additionally, the graph of FIG. 3 shows the results of numerical computations, with separate plots for the case where the correct answer is the first coherent state (white dots) and the case where the correct answer is the second coherent state (black dots).

[0052] In the example of FIG. 3, compared to FIG. 2, the average photon count is larger, and the noise is smaller. As a result, the example of FIG. 3 is considered to allow for higher accuracy in reading the bit value compared to the example of FIG. 2.

[0053] Control of quantum annealing is expected to enhance the accuracy of bit value readout by utilizing the readout qubit device **112** in a case where the photon count of the output signal from the computational qubit device **111** is low or the noise level is high.

[0054] Specifically, the output signal of the computational qubit device **111** is input to the readout qubit device **112**, and the interaction between these qubit devices, as well as the control of the readout qubit device **112**, are configured such that the readout qubit device **112** indicates the same bit value as the computational qubit device **111**.

[0055] The control of the computational qubit device **111** and the control of interactions among the computational qubit devices **111** are executed in accordance with the target problem of the quantum annealing. On the other hand, the control of the readout qubit device **112** and the interactions between the computational qubit device **111** and the readout qubit device **112** are not restricted to following by the target problem of the quantum annealing, offering increased flexibility. Thus, the control of the readout qubit device **112** and the control of the interactions therebetween can be configured to enable the output signal of the readout qubit device **112** to read bit values with greater accuracy than the output signal of the computational qubit device **111**.

[0056] Therefore, it is expected that reading the bit value from the readout qubit device **112** achieves greater accuracy compared to directly reading the bit value from the output of the computational qubit device **111**.

[0057] The coupler **120** allows multiple qubit devices **110** to interact with each other. The interaction of multiple qubit devices **110** is also referred to as the coupling of those qubit devices. [0058] Among couplers, a coupler that allows two qubit devices to interact with each other is also referred to as a two-qubit coupler.

[0059] One computational qubit device **111** and one readout qubit device **112** are coupled by a two-qubit coupler **121**. Then, the readout qubit device **112** and the two-qubit coupler **121** are controlled so that the readout qubit device **112** indicates the same bit value as the computational qubit device **111**.

[0060] The control device **200** controls the information processing device **100** and also reads out the results of computations performed by the information processing device **100**. The control device **200** may be implemented using a classical computer (of the von Neumann architecture).

[0061] The control unit **210** controls the information processing device **100**. The control unit **210** serves as an example of the control means.

[0062] The observation unit **220** reads out the results of the computations performed by the information processing device **100**.

[0063] FIG. **4** is a diagram showing a configuration example of the control device **200**. In the configuration shown in FIG. **4**, the control device **200** includes the control unit **210**, the observation unit **220**, a communication unit **231**, a display unit **232**, an operation input unit **233**, a storage unit **234**, and a processing unit **235**.

[0064] The communication unit **231**, the display unit **232**, the operation input unit **233**, the storage unit **234**, and the processing unit **235** may be configured using a computer (classical computer). The control device **200** may be configured as an integrated device, or may be configured as a combination of multiple devices.

[0065] The communication unit **231** communicates with the control unit **210** and the observation unit **220**. For example, the control unit **210** may be connected to the communication unit **231** via a signal line such as a bus. Then, based on the control signals received from the processing unit **235** through the communication unit **231**, the control unit **210** may generate control signals for each qubit device **110** and each coupler **120** using analog signals.

[0066] In addition, the control unit **210** and the observation unit **220** may be connected to each qubit device **110** and each coupler **120** via signal lines, such as coaxial cables for microwave signals. Then, the control unit **210** may transmit a control signal as an analog signal to each qubit device **110** and each coupler **120**. Moreover, the observation unit **220** may receive an output signal of the qubit device **110**. For example, the observation unit **220** may digitize an analog signal received from the qubit device **110**. The digital signal may then be received by the processing unit **235** via the communication unit **231**.

[0067] The display unit **232** includes a display screen such as a liquid crystal panel or a light emitting diode (LED) panel, and acquires various types of images. For example, the display unit **232** may display various information related to quantum annealing, such as the results of quantum annealing.

[0068] The operation input unit **233** includes input devices such as a keyboard and a mouse, and accepts user operations. For example, the operation input unit **233** may allow the user to make various quantum annealing settings, such as setting the number of execution repetitions for quantum annealing.

[0069] The storage unit **234** stores various types of data. For example, the storage unit **234** may store various quantum annealing data, such as the control schedule for the information processing device **100** and the results of individual quantum annealing executions.

[0070] The storage unit **234** is configured using a storage device included in the control device **200**.

[0071] The processing unit **235** controls each unit of the control device **200** and executes various processes. Functions of the processing unit **235** are executed by a central processing unit (CPU) included in the control device **200**, reading out a program from the storage unit **234** and executing

the program.

[0072] The control unit **210** controls the information processing device **100** as described above.

[0073] The observation unit **220** reads out the results of the computations performed by the information processing device **100**. The observation unit **220** then digitizes the acquired output signal of the readout qubit device **112**. The observation unit **220** or the processing unit **235** determines the bit value indicated by the readout qubit device **112** based on the digital signal.

[0074] Bit value determination based on digital signals can be performed, for example, through a process of calculating the I amplitude and Q amplitude from a time-series numerical array represented by the digital signals, and then a process of determining the bit value based on the I amplitude and Q amplitude. The observation unit **220** may perform these processes, or the processing unit **235** may perform these processes. Alternatively, the observation unit **220** may calculate the I amplitude and the Q amplitude, and the processing unit **235** may determine the bit value based on the I amplitude and the Q amplitude.

[0075] FIG. 5 is a diagram showing a first example of a combination of the computational qubit device **111** and the readout qubit device **112**. FIG. 5 shows an example in which a lumped-element Josephson parametric oscillator is used as the qubit device **110**. The lumped-element Josephson parametric oscillator is also referred to as a lumped-element qubit device **110a**.

[0076] In the example of FIG. 5, two lumped-element qubit devices **110a** are coupled by the two-qubit coupler **121** implemented using a capacitor **333**.

[0077] In the lumped-element qubit device **110a**, a loop is formed of a superconducting quantum interference device (SQUID) **310**, which consists of two Josephson junctions **311**, and a capacitor **330**. Moreover, an inductor (coil) **320** is provided near the superconducting quantum interference device **310** for inputting a pump signal by applying a magnetic field into the annular structure of the superconducting quantum interference device **310**.

[0078] In the loop formed of the superconducting quantum interference device **310** and the capacitor **330**, two terminals are provided at positions where the superconducting quantum interference device **310** and the capacitor **330** are connected in parallel. In the configuration of FIG. 5, the two terminals of both the computational qubit device **111** and the readout qubit device **112** are used as a terminal for signal input and a terminal for signal output. The signal input end is also referred to as input terminal **351**, and the signal output end is also referred to as output terminal **352**.

[0079] In the lumped-element qubit device **110a** serving as the computational qubit device **111**, the input terminal **351** is connected to the capacitor **331**, and a drive signal is input to the input terminal **351** via the capacitor **331**. The oscillation output is output externally via the capacitors **331**, **333**, which are connected to the input terminal **351** and the output terminal **352**, respectively.

[0080] Here, the signal output externally via the capacitor **331** connected to the input terminal **351** may be observed by the observation unit **220**. However, in such a case, it is possible that control for quantum annealing may result in the computational qubit device **111** not being in the state appropriate for reading out the bit value, which could lead to readout results that fail to distinguish the bit states, as shown in FIG. 2.

[0081] The oscillation signal output by the lumped-element qubit device **110a** serving as the computational qubit device **111** is also referred to as an oscillation output signal. The oscillation output signal is input to the lumped-element qubit device **110a** serving as the readout qubit device **112** via capacitor **333** serving as the two-qubit coupler **121**.

[0082] In the lumped-element qubit device **110a** serving as the readout qubit device **112**, the oscillation output signal from the computational qubit device **111** is input to the input terminal **351** of the readout qubit device **112** via the capacitor **333** serving as the two-qubit coupler **121**. The oscillation signal from the lumped-element qubit device **110a** serving as the readout qubit device **112** is output externally via the capacitors **333**, **332** connected to the input terminal **351** and the output terminal **352** of the readout qubit device **112**, respectively. Of these, the oscillation signal

that is output from the output terminal 352 of the readout qubit device 112 via the capacitor 332 to the control device 200 is also referred to as readout signal. Since the control of the lumped-element qubit device 110a serving as the readout qubit device 112 is not constrained by quantum annealing, it can be brought into a state suitable for reading, potentially resulting in a readout result that distinguishes the bit states, as shown in FIG. 3.

[0083] A Josephson parametric oscillator is a superconducting nonlinear resonator that includes a superconducting quantum interference device. In the present example embodiment, the superconducting quantum interference device is an annular-shaped structure including two Josephson junctions. In a Josephson parametric oscillator, in a case where microwaves having a frequency approximately twice the resonance frequency are applied to a superconducting quantum interference device, the magnetic flux passing through the annular structure is modulated at a frequency approximately twice the resonance frequency. As a result, the resonance frequency itself is subjected to parameter modulation at a frequency approximately twice the original resonance frequency, and the Josephson parametric oscillator oscillates parametrically at half the frequency of the applied microwave. The microwave applied at this time is referred to as a pump signal. The frequency of a pump signal is referred to as pump frequency.

[0084] The parametric oscillation performed by the Josephson parametric oscillator can take one of two oscillation states, either in phase or II out of phase, relative to the pump signal. By treating these two oscillation states as two levels representing bit values, the bit value can be represented by the oscillation state. These two oscillation states correspond to the two coherent states mentioned above.

[0085] In quantum computing using the Josephson parametric oscillator, the technique of adiabatic quantum computation is used. To adiabatically change the state of a qubit device, control parameter values must be set such that an adiabatic transition occurs from the initial state to the final state. To achieve this, the ground state before excitation must align with the ground state in the coordinate system after excitation.

[0086] Here, in the ground state before excitation, no pump signal is input, and the photon count in the Josephson parametric oscillator is zero.

[0087] Moreover, a rotating coordinate system is used as the coordinate system after excitation. The Hamiltonian $H/\hbar$ in the rotating coordinate system is expressed, for example, as Expression (1).

$$[00001] \quad H/\hbar = a^\dagger a - \frac{\chi}{2} a^\dagger a^\dagger a a + \left(a a + a^\dagger a^\dagger \right) \quad (1)$$

[0088] Δ is a value determined according to the frequency difference between the resonance frequency $\omega_{\text{sub.r}}$ of the Josephson parametric oscillator and half the pump frequency $\omega_{\text{sub.p}}$, and is expressed as Expression (2).

$$[00002] \quad \Delta = \omega_{\text{sub.r}} - \frac{\omega_{\text{sub.p}}}{2} \quad (2)$$

[0089] The symbol “a” with a dagger (†) superscript denotes a creation operator.

[0090] The symbol “a” denotes an annihilation operator.

[0091] The symbol χ denotes a nonlinearity.

[0092] The symbol β denotes a pump signal intensity. The pump signal intensity is also referred to as pump intensity.

[0093] If the photon count in the Josephson parametric oscillator is n , then in a case where $\beta=0$, the energy is given by $n\Delta - n(n-1)\chi/2$. Here, n is an integer where $n \geq 0$. In general, in a case where $\chi > 0$ and $\Delta < 0$, the state with zero photons, which corresponds to the ground state for $\beta=0$, becomes the maximum energy state. In other words, if $\Delta < 0$, the energy is maximized in a case where $n=0$ among $n=0, 1, 2, \dots$

[0094] In a case where $\Delta < 0$ in Expression (2), $\omega_{\text{sub.p}} > 2\omega_{\text{sub.r}}$.

[0095] The energy ground state after parametric oscillation is a coherent state, and in the rotating coordinate system, this coherent state represents the state with the maximum energy. Therefore, to

adiabatically change the state of the Josephson parametric oscillator, it is necessary to ensure that the state with zero photons is the energy maximum state in a case where $\beta=0$. In other words, the pump frequency $\omega_{\text{sub.p}}$ should be set to a value greater than twice the resonant frequency $\omega_{\text{sub.r}}$ of the Josephson parametric oscillator. In this case, by gradually changing the pump signal intensity β from 0 to a finite value, the state of the Josephson parametric oscillator can be adiabatically evolved, and it is expected that the state corresponding to the optimal solution will be obtained. [0096] On the other hand, considering the case where the initial state is not the ground state, a relaxation process involving interactions with photon exchanges must occur before reaching the state corresponding to the optimal solution. Therefore, if the initial state is not the ground state, it is expected that time will be required before reaching the state corresponding to the optimal solution. Additionally, if the initial state is not the ground state, transitions to non-optimal states may also occur simultaneously.

[0097] In this way, it is preferable to set the pump frequency $\omega_{\text{sub.p}}$ to a value greater than twice the resonant frequency $\omega_{\text{sub.r}}$ of the Josephson parametric oscillator, in order to adiabatically change the state of the Josephson parametric oscillator.

[0098] In contrast, in a case where the pump frequency $\omega_{\text{sub.p}}$ is set to a value greater than twice the resonant frequency $\omega_{\text{sub.r}}$ of the Josephson parametric oscillator, the intensity of the output oscillation signal output by the Josephson parametric oscillator becomes relatively weaker. Here, the Josephson parametric oscillator, after excitation, enters the coherent state, and the average photon count N in the coherent state is approximately given by Expression (3).

$$[00003] \ N = (2 + \Delta) / \chi \quad (3)$$

[0099] In Expression (3), the larger the value of Δ , the greater the average photon count N , and the stronger the intensity of the oscillation signal becomes. In a case where $\Delta < 0$, that is, in a case where the pump frequency $\omega_{\text{sub.p}}$ is greater than twice the resonance frequency $\omega_{\text{sub.r}}$ of the Josephson parametric oscillator, the intensity of the oscillation signal is weaker than in a case where $\Delta \geq 0$. It is considered that the accuracy of reading out the bit value decreases if the intensity of the oscillation signal of the Josephson parametric oscillator cannot be made sufficiently strong.

[0100] In addition to increasing the value of Δ in Expression (3) to increase the average photon count N , it is also possible to increase the pump signal intensity β and reduce the nonlinearity χ . However, both β and χ have upper and lower limit values that are determined by the experimental conditions. Therefore, increasing the value of the pump signal intensity β and decreasing the value of the nonlinearity χ imposes limitations on increasing the average photon count.

[0101] Additionally, in a case where reading the bit value from the oscillation signal of the Josephson parametric oscillator, amplifying the oscillation signal with an amplifier before reading the bit value may be considered. However, amplifying the oscillation signal results in the amplification of the noise contained within the signal as well. As shown in the example of FIG. 2, in a case where the average photon count is relatively small, it may be difficult to distinguish between the two coherent states, even with use of the amplified oscillation signal.

[0102] In order to accurately read the bit value from the oscillation signal of the Josephson parametric oscillator, it is necessary for the average photon count in the oscillation signal before amplification to be sufficiently large, as shown in the example of FIG. 3, and for the intensity of the oscillation signal to be sufficiently high relative to the noise level.

[0103] Therefore, the information processing device **100** is provided with the readout qubit device **112**. The observation unit **220** reads the bit value from the readout qubit device **112** rather than directly from the computational qubit device **111**.

[0104] As mentioned above, the readout qubit device **112** is a qubit device **110** that is not directly used in bit value computation through quantum computing, and provides greater flexibility in control compared to the computational qubit device **111**. As a result, the control unit **210** can perform control better suited for bit value readout on the readout qubit device **112**, which is

expected to enable bit value reading with relatively high accuracy.

[0105] For example, in a case where the qubit device **110** is configured using a Josephson parametric oscillator as in the example of FIG. 5, the pump frequency $\omega_{\text{sub.p}}$ for the computational qubit device **111** may be set to a value greater than twice the resonance frequency $\omega_{\text{sub.r}}$ of the Josephson parametric oscillator. As a result, the control unit **210** can adiabatically change the state of the computational qubit device **111**.

[0106] On the other hand, for the readout qubit device **112**, the pump frequency $\omega_{\text{sub.p}}$ may be set to a value less than or equal to twice the resonance frequency $\omega_{\text{sub.r}}$ of the Josephson parametric oscillator. As a result, the intensity of the readout signal (the oscillation signal output by the readout qubit device **112**) can be increased, enabling the observation unit **220** to read out the bit value with relatively high accuracy.

[0107] Additionally, since there is no need to change the state of the readout qubit device **112** adiabatically, the value of the nonlinearity parameter χ can be reduced compared to that of the computational qubit device **111**. In this respect also, the intensity of the readout signal can be increased, allowing the observation unit **220** to read out the bit value with relatively high accuracy.

[0108] For the two-qubit-coupler, two Josephson parametric oscillators are connected by a capacitor. By inputting pump signals of the same frequency to these two Josephson parametric oscillators, they can be two-qubit-coupled. In the example of FIG. 5, pump signals of the same frequency are input to the computational qubit device **111** and the readout qubit device **112**. It should be noted that the term “same frequency” here does not strictly mean exactly the same frequency. It is sufficient if the frequencies are close enough to ensure that the phase difference does not exceed $\pi/2$ during the readout time.

[0109] As for the resonance frequency, as mentioned above, for the computational qubit device **111**, it is possible to set the pump frequency $\omega_{\text{sub.p}}$ to a value greater than twice the resonance frequency $\omega_{\text{sub.r}}$ of the Josephson parametric oscillator. For the readout qubit device **112**, the pump frequency $\omega_{\text{sub.p}}$ can be set to a value equal to or less than twice the resonance frequency $\omega_{\text{sub.r}}$ of the Josephson parametric oscillator.

[0110] Thus, the following relationship may be satisfied: the resonance frequency of the computational qubit device **111** is less than half the common pump frequency for the computational qubit device **111** and the readout qubit device **112**, which is, in turn, less than the resonance frequency of the readout qubit device **112**.

[0111] The strength of the two-qubit-coupling between the two Josephson parametric oscillators can be varied by the phase difference of the pump signals of these two Josephson parametric oscillators. Accordingly, the control unit **210** may adjust the relative phase of the pump signal for the readout qubit device **112** with respect to the pump signal for the computational qubit device **111**, so that the strength of the two-qubit-coupling between the computational qubit device **111** and the readout qubit device **112** becomes as strong as possible.

[0112] By two-qubit-coupling the computational qubit device **111** and the readout qubit device **112**, a correlation occurs between the oscillation phases of these two qubit devices **110**. As a result, the readout qubit device **112** is expected to indicate the same bit value as that indicated by the computational qubit device **111**.

[0113] In particular, since the readout qubit device **112** is two-qubit-coupled to the computational qubit device **111**, unlike the case of four-qubit coupling, the readout qubit device **112** interacts only with this computational qubit device **111**. In this respect, the oscillation phase of the readout qubit device **112** can be directly reflected by the oscillation phase of the computational qubit device **111**.

[0114] Here, if the oscillation phase of the computational qubit device **111** is influenced by the oscillation phase of the readout qubit device **112** due to the two-qubit-coupling, the computation accuracy of the bit value by quantum computing may be reduced. In particular, it is considered that the oscillation phase of the computational qubit device **111** is more likely to be influenced by the oscillation phase of the readout qubit device **112** in a case where the oscillation signal intensity of

the readout qubit device **112** is higher than that of the computational qubit device **111**.

[0115] To avoid or mitigate such influence, the control unit **210** may be configured to initiate the excitation of the readout qubit device **112** after the state transition of the computational qubit device **111** has been completed and the computational qubit device **111** has reached the coherent state. That is to say, the control unit **210** may be configured to initiate inputting the pump signal to the readout qubit device **112** and transition the state of the readout qubit device **112** after the state transition of the computational qubit device **111** has been completed.

[0116] In the example shown in FIG. 5, the input of the drive signal and the output of the oscillation output signal are performed at different terminals of the computational qubit device **111**. This eliminates the need for a mechanism, such as a circulator, to separate the output signal and the input signal, compared to the case where the drive signal is input and the oscillation output signal is output at the same terminal of the Josephson parametric oscillator.

[0117] Furthermore, the computational qubit device **111** can, in a state receiving the drive signal input, output the oscillation output signal, which can be input to the readout qubit device **112** via the two-qubit coupler **121**. This enables the stabilization of the oscillation output signal from the computational qubit device **111**, which is expected to improve the accuracy of bit value reading performed by the observation unit **220**.

[0118] Moreover, in the example shown in FIG. 5, the input of the oscillation output signal from the computational qubit device **111** and the output of the readout signal are performed at different terminals of the readout qubit device **112**. This eliminates the need for a mechanism, such as a circulator, to separate the output signal and the input signal, compared to the case where the drive signal is input and the oscillation output signal is output at the same terminal of the Josephson parametric oscillator.

[0119] Also, the readout qubit device **112** can output the readout signal while receiving an input of the oscillation output signal. This enables the stabilization of the readout signal output from the readout qubit device **112**, which is expected to improve the accuracy of bit value reading performed by the observation unit **220**.

[0120] FIG. 6 is a diagram showing a second example of a combination of the computational qubit device **111** and the readout qubit device **112**. FIG. 6 shows an example in which a distributed-element Josephson parametric oscillator is used as the qubit device **110**. The distributed-element Josephson parametric oscillator is also referred to as distributed-element qubit device **110b**.

[0121] In the example of FIG. 6, two distributed-element qubit devices **110b** are coupled by the two-qubit coupler **121** implemented using a capacitor **333**.

[0122] In the distributed-element qubit device **110b**, a superconducting quantum interference device **310**, which has an annular structure with two Josephson junctions **311**, is positioned between two quarter-wavelength ($\lambda/4$) resonators **340**. The quarter-wavelength resonator **340**, the superconducting quantum interference device **310**, and the quarter-wavelength resonator **340** are connected in series in this order. Also, an inductor **320** for inputting a pump signal is provided near one of the two Josephson junctions.

[0123] Of the two terminals of the quarter-wavelength resonators **340**, the terminal that is not connected to the superconducting quantum interference device **310** corresponds to the terminal of the distributed-element qubit device **110b**.

[0124] In the distributed-element qubit device **110b** serving as the computational qubit device **111**, the input terminal **351** is connected to the capacitor **331**, and a drive signal is input to the input terminal **351** via the capacitor **331**. Moreover, an oscillation signal is output from the output terminal **352**.

[0125] The oscillation signal output by the distributed element qubit device **110b** serving as the computational qubit device **111** is also referred to as oscillation output signal. The oscillation output signal is input to the distributed-element qubit device **110b** serving as the readout qubit device **112** via capacitor **333** serving as the two-qubit coupler **121**.

[0126] In the distributed element qubit device **110b** serving as the readout qubit device **112**, the oscillation output signal is input to the input terminal **351** via the capacitor **333** serving as the two-qubit coupler **121**. Moreover, the output terminal **352** is connected to the capacitor **332**, and the oscillation signal is output from the output terminal **352** to the control device **200** via the capacitor **332**. The oscillation signal output by the distributed-element qubit device **110b** serving as the readout qubit device **112** is also referred to as readout signal.

[0127] Also in the case where the distributed-element qubit device **110b** is used as a qubit device **110**, the pump frequency $\omega_{\text{sub.p}}$ for the computational qubit device **111** may be set to a value greater than twice the resonance frequency $\omega_{\text{sub.r}}$ of the Josephson parametric oscillator. As a result, the control unit **210** can adiabatically change the state of the computational qubit device **111**.

[0128] Moreover, for the readout qubit device **112**, the pump frequency $\omega_{\text{sub.p}}$ may be set to a value less than or equal to twice the resonance frequency $\omega_{\text{sub.r}}$ of the Josephson parametric oscillator. As a result, the intensity of the readout signal (the oscillation signal output by the readout qubit device **112**) can be increased, enabling the observation unit **220** to read out the bit value with relatively high accuracy.

[0129] Also in the example of FIG. 6, since there is no need to change the state of the readout qubit device **112** adiabatically, the value of the nonlinearity parameter χ can be reduced compared to that of the computational qubit device **111**. In this respect also, the intensity of the readout signal can be increased, allowing the observation unit **220** to read out the bit value with relatively high accuracy.

[0130] Also in the example of FIG. 6, by inputting a pump signal of the same frequency to both the computational qubit device **111** and the readout qubit device **112**, it is possible to two-qubit-couple the computational qubit device **111** and the readout qubit device **112**. It should be noted that the term “same frequency” here does not strictly mean exactly the same frequency. It is sufficient if the frequencies are close enough to ensure that the phase difference does not exceed $\pi/2$ during the readout time.

[0131] Also in the example of FIG. 6, the control unit **210** may adjust the relative phase of the pump signal for the readout qubit device **112** with respect to the pump signal for the computational qubit device **111**, so that the strength of the two-qubit-coupling between the computational qubit device **111** and the readout qubit device **112** becomes as strong as possible.

[0132] Also in the example of FIG. 6, by two-qubit-coupling the computational qubit device **111** and the readout qubit device **112**, a correlation occurs between the oscillation phases of these two qubit devices **110**. As a result, the readout qubit device **112** is expected to indicate the same bit value as that indicated by the computational qubit device **111**.

[0133] In particular, since the readout qubit device **112** is two-qubit-coupled to the computational qubit device **111**, unlike the case of four-qubit coupling, the readout qubit device **112** interacts only with this computational qubit device **111**. In this respect, the oscillation phase of the readout qubit device **112** can be directly reflected by the oscillation phase of the computational qubit device **111**.

[0134] Also in the example of FIG. 6, the control unit **210** may be configured to initiate the excitation of the readout qubit device **112** after the state transition of the computational qubit device **111** has been completed and the computational qubit device **111** has reached the coherent state. That is to say, the control unit **210** may be configured to initiate inputting the pump signal to the readout qubit device **112** and transition the state of the readout qubit device **112** after the state transition of the computational qubit device **111** has been completed.

[0135] Also in the example shown in FIG. 6, the input of the drive signal and the output of the oscillation output signal are performed at different terminals of the computational qubit device **111**. This eliminates the need for a mechanism, such as a circulator, to separate the output signal and the input signal, compared to the case where the drive signal is input and the oscillation output signal is output at the same terminal of the Josephson parametric oscillator.

[0136] Also in the example of FIG. 6, the computational qubit device **111** can, in a state receiving the drive signal input, output the oscillation output signal, which can be input to the readout qubit

device **112** via the two-qubit coupler **121**. This enables the stabilization of the oscillation output signal from the computational qubit device **111**, which is expected to improve the accuracy of bit value reading performed by the observation unit **220**.

[0137] Also in the example of FIG. **6**, the input of the oscillation output signal from the computational qubit device **111** and the output of the readout signal are performed at different terminals of the readout qubit device **112**. This eliminates the need for a mechanism, such as a circulator, to separate the output signal and the input signal, compared to the case where the drive signal is input and the oscillation output signal is output at the same terminal of the Josephson parametric oscillator.

[0138] Also in the example of FIG. **6**, the readout qubit device **112** can output the readout signal while receiving an input of the oscillation output signal. This enables the stabilization of the readout signal output from the readout qubit device **112**, which is expected to improve the accuracy of bit value reading performed by the observation unit **220**.

[0139] FIG. **7** is a diagram showing a third example of a combination of the computational qubit device **111** and the readout qubit device **112**. FIG. **7** shows another example in which a lumped-element Josephson parametric oscillator is used as the qubit device **110**.

[0140] In the example of FIG. **7**, the manner in which the loop formed of the superconducting quantum interference device **310** and the capacitor **330** is set up differs from that in FIG. **5**.

[0141] In FIG. **7**, for both the computational qubit device **111** and the readout qubit device **112**, one of the two terminals, where the superconducting quantum interference device **310** and the capacitor **330** are arranged in a parallel connection, is connected to the signal path. The other of the two terminals is grounded (GND).

[0142] In other respects, the example of FIG. **7** is similar to that in the case of FIG. **5**.

[0143] Thus, it is possible for only one of the two terminals of the loop between the superconducting quantum interference device **310** and the capacitor **330** to be connected to the signal path. This terminal is also referred to as input/output terminal **353**.

[0144] It is also possible for a lumped-element Josephson parametric oscillator and a distributed-element Josephson parametric oscillator to be used together. For example, a lumped-element qubit device **110a** may be used as the computational qubit device **111**, and a distributed-element qubit device **110b** may be used as the readout qubit device **112**. Alternatively, the distributed-element qubit device **110b** may be used as the computational qubit device **111**, while the lumped-element qubit device **110a** may be used as the readout qubit device **112**.

[0145] FIG. **8** is a diagram showing an example of control timing for the computational qubit device **111** and the readout qubit device **112**.

[0146] In FIG. **8**, graphs G1 to G4 are shown aligned in time. The graph G1 is a graph showing the input timing of a drive signal to the computational qubit device **111**. The graph G2 is a graph showing the input timing of a pump signal to the computational qubit device **111**. The graph G3 is a graph showing the input timing of a pump signal to the readout qubit device **112**. The graph G4 is a graph showing the read timing of a readout signal from the readout qubit device **112**.

[0147] The horizontal axis of each graph represents the time elapsed from a predetermined reference time.

[0148] The vertical axis of graphs G1 to G3 represents the intensity of the input signal to the qubit device **110**, normalized to a value within the range of 0 to 1. The value 0 on the vertical axis indicates the signal intensity in a case where no signal is input to the qubit device **110**. The value 1 on the vertical axis indicates the signal intensity that is set as a steady-state value.

[0149] The vertical axis of the graph G4 represents the distinction between “ON” and “OFF” of the readout period. The readout period here refers to a temporal period during which an oscillation signal from the qubit device **110** is read out. The readout period being “ON” indicates a timing in a case where the observation unit **220** is reading out the oscillation signal. The readout period being “OFF” indicates a timing in a case where the observation unit **220** is not reading out the oscillation

signal.

[0150] In the example of FIG. 8, the control unit **210** starts inputting the drive signal and pump signal to the computational qubit device **111** at time **T1**, and at time **T2**, the intensity of these signals is set to the steady-state value. Before time **T1**, in a case where the pump signal is not being input, the state of the computational qubit device **111** is in its initial state. After time **T2**, in a case where the intensity of the pump signal has reached the steady-state value, the state of the computational qubit device **111** becomes the coherent state (that is, a state in which the qubit value is determined as being one of two binary values).

[0151] Moreover, the control unit **210** starts the input of the pump signal to the readout qubit device **112** at time **T3**, after time **T2**, and at time **T4**, the intensity of the pump signal is set to the steady-state value. Before time **T3**, in a case where the pump signal is not being input, the state of the readout qubit device **112** is in its initial state. After time **T4**, in a case where the intensity of the pump signal has reached the steady-state value, the state of the readout qubit device **112** becomes the coherent state.

[0152] The observation unit **220** reads the readout signal (the oscillation signal from the readout qubit device **112**) during the readout period, which spans from time **T5** to **T6**, after the pump signal intensity of the readout qubit device **112** has reached the steady-state value at time **T4**.

[0153] After time **T6**, from time **T7** to time **T8**, the control unit **210** reduces the intensity of the pump signal to the readout qubit device **112**, and at time **T8**, the input of the pump signal to the readout qubit device **112** is terminated.

[0154] Thus, in the example of FIG. 8, the control unit **210** can input the pump signal to the readout qubit device **112** and initiate excitation after the time in a case where the computational qubit device **111** has reached the coherent state. This makes it possible to prevent or mitigate the influence of the oscillation phase of the computational qubit device **111** on the oscillation phase of the readout qubit device **112**, as described above.

[0155] Moreover, by inputting a pump signal such that $\Delta > 0$ to the Josephson parametric oscillator and inputting a drive signal, the Josephson parametric oscillator can be excited in synchronization with the drive signal.

[0156] Therefore, after the computational qubit device **111** reaches the coherent state, the control unit **210** may input a pump signal to the readout qubit device **112** such that $\Delta > 0$. As a result, the readout qubit device **112** is expected to oscillate in the same phase as the oscillation phase of the computational qubit device **111**, allowing the observation unit **220** to read out the bit value represented by the computational qubit device **111** from the readout qubit device **112**.

[0157] Additionally, the condition $\Delta > 0$ corresponds to the pump frequency $\omega_{\text{sub.p}}$ being less than twice the resonance frequency $\omega_{\text{sub.r}}$ of the Josephson parametric oscillator. Therefore, as described above, the intensity of the oscillation signal from the readout qubit device **112** becomes relatively high, which is expected to enable the observation unit **220** to read out the bit value indicated by the readout qubit with relatively high accuracy.

[0158] FIG. 9 is a diagram showing an example of a procedure of processing performed by the control device **200**. FIG. 9 shows an example of the procedure of processing performed by the control device **200** during a single solution search in quantum annealing. The control device **200** repeats the process shown in FIG. 9 for each iteration of the solution search in quantum annealing.

[0159] In the process shown in FIG. 9, the control unit **210** controls the computational qubit device **111** (Step **S11**). Specifically, the control unit **210** inputs control signals, such as a pump signal, to the computational qubit device **111** through the communication unit **231**.

[0160] After the state of the computational qubit device **111** reaches the coherent state, the control unit **210** performs control over the readout qubit device **112** (Step **S12**). Specifically, the control signals, such as a pump signal, are input from the control unit **210** to the readout qubit device **112**.

[0161] After the state of the readout qubit device **112** reaches the coherent state, the observation unit **220** reads out the bit value from the readout qubit device **112** (Step **S13**). Specifically, the

observation unit **220** estimates the bit value by measuring the phase of the readout signal, which is the oscillation signal output by the readout qubit device **112**.

[0162] After Step **S13**, the control device **200** ends the process of FIG. **9**. The term “end” here refers to the termination of the output from the control unit **210** and the measurement of the readout signal performed by the observation unit **220**.

[0163] In the example provided here, the information processing qubit is described as a computational qubit. However, the information processing qubit may also be used for purposes other than computation. As described above for computational qubit devices, a computational qubit is a qubit that is used to perform a computation. An information processing qubit is a quantum bit that is used widely for information processing, not limited to computations. For example, information processing qubits may be used as storage qubits.

[0164] As described above, the readout qubit device **112** is coupled by the coupler **120** to the information processing qubit device so as to assume the same bit value as the information processing qubit device.

[0165] In the information processing device **100**, the control of the readout qubit device **112** does not need to be the same as that for an information processing qubit device, and the degree of freedom for controlling the readout qubit device **112** is relatively high. As a result, it is expected that the information processing device **100** will be able to control the readout qubit device **112** in a manner more suitable for reading out bit values, thereby making it possible to read out bit values with higher accuracy.

[0166] Moreover, the information processing qubit device is a computational qubit device **111**.

[0167] In the information processing device **100**, under the control of the control device **200**, computations such as quantum annealing can be performed.

[0168] Also, the readout qubit device **112** and the computational qubit device **111** coupled to each other are excited at the same frequency. It should be noted that the term “same frequency” here does not strictly mean exactly the same frequency. It is sufficient if the frequencies are close enough to ensure that the phase difference does not exceed $\pi/2$ during the readout time.

[0169] As a result, in the information processing device **100**, these two qubit devices **110** can be coupled such that the readout qubit device **112** has the same bit value as the computational qubit device **111**. As a result, it is expected that in the information processing device **100**, the same bit value as the bit value of the computational qubit device **111** can be read out from the readout qubit device **112**.

[0170] Furthermore, the pump frequency of the computational qubit device **111** is higher than twice the resonance frequency. The pump frequency of the readout qubit device **112** is lower than twice the resonance frequency.

[0171] In the information processing device **100**, the pump frequency of the computational qubit device **111** is higher than twice the resonance frequency, so that the state of the computational qubit device **111** can be changed adiabatically. Moreover, by making the pump frequency of the readout qubit device **112** lower than twice the resonance frequency, the intensity of the oscillation signal of the readout qubit device **112** can be made relatively high. In this respect, according to the information processing device **100**, it is expected that the bit value can be read out with relatively high accuracy.

[0172] Also, the readout qubit device **112** and the computational qubit device **111** coupled to each other are excited at the same pump frequency. The resonance frequency of the computational qubit device **111** is lower than one-half the pump frequency, and the resonance frequency of the readout qubit device **112** is higher than one-half the pump frequency.

[0173] In the information processing device **100**, the readout qubit device **112** and the computational qubit device **111** are excited at the same pump frequency, so that these two qubit devices **110** can be two-qubit coupled so as to take the same value. As a result, it is expected that in the information processing device **100**, the same bit value as the bit value of the computational

qubit device **111** can be read out from the readout qubit device **112**.

[0174] In the information processing device **100**, the resonance frequency of the computational qubit device **111** is lower than one-half the pump frequency, so that the state of the computational qubit device **111** can be changed adiabatically. Moreover, since the resonance frequency of the readout qubit device **112** is higher than one-half the pump frequency, the oscillation signal phase of the readout qubit device **112** can be synchronized with that of the computational qubit device **111**, and the intensity of the oscillation signal from the readout qubit device **112** can be made relatively high. In this respect, according to the information processing device **100**, it is expected that the bit value can be read out with relatively high accuracy.

[0175] Furthermore, the control unit **210** can excite the readout qubit device **112** after the computational qubit device **111** reaches the coherent state.

[0176] In this regard, it is expected that the information processing device **100** will make it easier to control the readout qubit device **112** so that the bit value of the readout qubit device **112** becomes the same value as the bit value of the computational qubit device **111**. According to the information processing device **100**, in this respect, it is expected that the same bit value as the bit value of the computational qubit device **111** can be read out from the readout qubit device **112**.

Second Example Embodiment

[0177] In a second example embodiment, a first example of applying the information processing system **1** to a model based on the Lechner-Hauke-Zoller (LHZ) scheme is described. A model based on the LHZ scheme is also referred to as an LHZ model.

[0178] In the second example embodiment, the configuration of the information processing device **100** is concretized from the configuration of the first example embodiment. In other respects, the information processing system **1** of the second example embodiment is similar to that of the first example embodiment.

[0179] FIG. **10** is a diagram showing the first example of the configuration of the information processing device **100** where an LHZ model is used. In the example of FIG. **10**, the information processing device **100** includes eight computational qubit devices **111**, three readout qubit devices **112**, three two-qubit couplers **121**, and three four-qubit couplers **122**. Just as the two-qubit coupler **121** corresponds to an example of the coupler **120**, the four-qubit coupler **122** also corresponds to an example of the coupler **120**.

[0180] The configuration of the information processing device **100** shown in FIG. **10** corresponds to a specific example of the configuration of the information processing device **100** shown in FIG. **1**.

[0181] In a case where distinguishing the eight computational qubit devices **111** from one another, they may be referred to as computational qubit devices **111-1**, **111-2**, . . . , **111-8**. In a case where distinguishing the three readout qubit devices **112** from one another, they may be referred to as readout qubit devices **112-1**, **112-2**, and **112-3**. In a case where distinguishing the three two-qubit couplers **121** from one another, they may be referred to as two-qubit couplers **121-1**, **121-2**, and **121-3**. In a case where distinguishing the three four-qubit couplers **122** from one another, they may be referred to as four-qubit couplers **122-1**, **122-2**, and **122-3**.

[0182] FIG. **10** shows an example in which the readout qubit devices **112** are provided in a physical model obtained by converting a fully connected 4-bit logical model using the LHZ scheme. However, the information processing device **100** is applicable to problems represented by logical models of various bit sizes, not just limited to 4 bits.

[0183] The bit value of each of the computational qubit devices **111-1** through **111-6** represents the product of two logical bits. In the representation of these computational qubit devices **111**, the two digits shown in a circle indicate the identification numbers of the logical bits that are being multiplied. For example, the bit value of the computational qubit device **111-1** indicates the product of the first logical bit and the fourth logical bit.

[0184] The computational qubit devices **111-7** and **111-8** are controlled such that their bit values are

fixed at +1.

[0185] Each of the four-qubit couplers **122** couples four computational qubit devices **111**. In the representation of the four-qubit coupler **122**, the digit “4” shown in a square indicates that the coupling count (the number of qubit devices to which the coupler couples) is four.

[0186] Among the computational qubit devices **111-1** to **111-6**, the computational qubit devices **111-4** to **111-6** are targets for bit value reading. However, the observation unit **220** reads the bit value from the readout qubit device **112** rather than directly from the computational qubit device **111**.

[0187] The configuration in FIG. **10** can be interpreted as a configuration in which the computational qubit devices **111** and the four-qubit couplers **122** form a model based on the LHZ scheme, with the readout qubit devices **112** being incorporated via the two-qubit couplers **121**.

[0188] Furthermore, each of the multiple qubit devices **110** is configured using a nonlinear oscillator. Of the readout qubit device **112** and the computational qubit device **111** coupled to each other, it is considered that the nonlinearity of the readout qubit device **112** is made lower than that of the computational qubit device **111**.

[0189] According to the information processing device **100**, since the nonlinearity of the readout qubit device **112** is lower than the nonlinearity of the computational qubit device **111**, it is expected that the intensity of the output signal of the readout qubit device **112** can be made higher than the intensity of the output signal of the computational qubit device **111**. According to the information processing device **100**, in this respect, it is expected that the bit value can be read out with higher accuracy compared to directly reading out the bit value from the computational qubit device **111**.

[0190] Each of the readout qubit devices **112** is coupled to a corresponding computational qubit device **111** by the two-qubit coupler **121**. The readout qubit device **112** is controlled so as to display the same bit value as the computational qubit device **111** to which it is coupled. Specifically, the readout qubit device **112-1** is coupled to the computational qubit device **111-4** via the two-qubit coupler **121-1** and is controlled so as to display the same bit value as the computational qubit device **111-4**. The readout qubit device **112-2** is coupled to the computational qubit device **111-5** via the two-qubit coupler **121-2** and is controlled so as to display the same bit value as the computational qubit device **111-5**. The readout qubit device **112-3** is coupled to the computational qubit device **111-6** via the two-qubit coupler **121-3** and is controlled so as to display the same bit value as the computational qubit device **111-6**.

[0191] In the representation of the two-qubit coupler **121**, the digit “2” shown in a square indicates that the coupling count is two.

[0192] The observation unit **220** reads out the bit values from the readout qubit devices **112-1** to **112-3**, thereby reading out the bit values from the computational qubit devices **111-4** to **111-6**.

[0193] In a case where performing quantum annealing using the LHZ scheme with a Josephson parametric oscillator, the state is adiabatically changed until reaching the coherent state, while simultaneously inputting a pump signal to the Josephson parametric oscillator to induce interactions.

[0194] In the example shown in FIG. **10**, the control unit **210** adiabatically changes the state until it reaches the coherent state, while simultaneously inputting a pump signal to the computational qubit device **111** to induce interactions. After the computational qubit device **111** reaches the coherent state, the control unit **210** inputs a pump signal to the readout qubit device **112**, inducing two-qubit interactions with the computational qubit device **111**. Then, the observation unit **220** reads out the bit value from the readout qubit device **112**.

[0195] In the second example embodiment also, the information processing qubit is described as a computational qubit. However, the information processing qubit may also be used for purposes other than computation.

[0196] As described above, the computational qubit device **111** and one or more four-qubit couplers **122** among the couplers **120** form an LHZ scheme model.

[0197] In this respect, according to the information processing device **100**, it is expected that the bit value can be read out with relatively high accuracy from the LHZ scheme model. In this respect, it is expected that the information processing device **100** will be able to calculate with higher accuracy a solution to a problem represented by a fully connected logical model.

[0198] In particular, in the information processing device **100**, the control of the readout qubit device **112** does not need to be the same as that for the computational qubit device **111**, and the degree of freedom for controlling the readout qubit device **112** is relatively high. As a result, it is expected that the information processing device **100** will be able to control the readout qubit device **112** in a manner more suitable for reading out bit values, thereby making it possible to read out bit values with higher accuracy.

Third Example Embodiment

[0199] In a third example embodiment, a second example in which the information processing system **1** is applied to an LHZ model will be described.

[0200] In the third example embodiment, the configuration of the information processing device **100** is concretized from the configuration of the first example embodiment. In other respects, the information processing system **1** of the third example embodiment is similar to that of the first example embodiment.

[0201] FIG. **11** is a diagram showing the second example of the configuration of the information processing device **100** where an LHZ model is used. In the example of FIG.

[0202] **11**, the information processing device **100** includes twenty-two qubit devices **110** and thirteen four-qubit couplers **122**. Eight of the twenty two qubit devices **110** are used as computational qubit devices **111**. The arrangement and application of the qubit devices **110** used as the computational qubit devices **111** are the same as those in FIG. **10**. Similarly to FIG. **10**, the computational qubit devices are denoted as **111-1**, **111-2**, . . . , **111-8**.

[0203] Of the remaining fourteen qubit devices **110**, three are used as readout qubit devices **112**. The arrangement and application of the qubit devices **110** used as the readout qubit devices **112** are the same as those in FIG. **10**. Similarly to FIG. **10**, the readout qubit devices are denoted as **112-1**, **112-2**, . . . , **112-3**.

[0204] The remaining eleven qubit devices **110** are unused.

[0205] In a case where distinguishing the four-qubit couplers **122** in the example of

[0206] FIG. **11** from one another, they may be referred to as four-qubit couplers **122-1**, **122-2**, . . . , **122-13**.

[0207] The four-qubit couplers **122-1** to **122-3**, as in the case of FIG. **10**, couple four computational qubit devices **111**.

[0208] On the other hand, the four-qubit couplers **122-4** to **122-6**, as in the case of the two-qubit couplers **121-1** to **121-3** in FIG. **10**, each couple a single computational qubit device **111** with a single readout qubit device **112**.

[0209] The four-qubit couplers **122-7** to **122-13** are unused.

[0210] The control unit **210** may be configured not to parametrically excite the unused qubit devices **110**.

[0211] In the four-qubit coupling of the Josephson parametric oscillators, the pump frequencies of the four Josephson parametric oscillators are configured to satisfy the relationship of Expression (4).

[00004]
$$\omega_{p,1} + \omega_{p,2} = \omega_{p,3} + \omega_{p,4} \quad (4)$$

[0212] $\omega_{\text{sub.p},1}$, $\omega_{\text{sub.p},2}$, $\omega_{\text{sub.p},3}$, and $\omega_{\text{sub.p},4}$ represent the pump frequencies of the four Josephson parametric oscillators that are four-qubit-coupled.

[0213] In the case of a four-qubit-coupling, the pump frequencies $\omega_{\text{sub.p},1}$, $\omega_{\text{sub.p},2}$, $\omega_{\text{sub.p},3}$, and $\omega_{\text{sub.p},4}$ of the four Josephson parametric oscillators to be coupled are set to distinct values to prevent the occurrence of two-qubit-coupling.

[0214] In the example shown in FIG. **11**, for example, the four-qubit coupler **122-1** couples the four computational qubit devices **111-1**, **111-2**, **111-3**, and **111-5**. The control unit **210** sets the frequencies of the pump signals input to these four computational qubit devices **111** to distinct values, ensuring that the relationship of Expression (4) is satisfied.

[0215] On the other hand, in a case where using a four-qubit coupler to two-qubit-couple two Josephson parametric oscillators, the pump frequencies of the two Josephson parametric oscillators are set to be identical. In the example shown in FIG. **11**, for example, the control unit **210** sets the frequency of the pump signal input to the computational qubit device **111** and the frequency of the pump signal input to the readout qubit device to the same value, so that the four-qubit coupler **122-4** two-qubit-couples the computational qubit device **111** and the readout qubit device.

[0216] It is conceivable that the resonance frequencies of the two qubit devices **110** that are not coupled are sufficiently detuned from those of the two qubit devices **110** that are coupled. For example, the resonance frequency can be adjusted by inputting a direct current to the inductor **320** and changing the magnetic flux passing through the annular structure of the superconducting quantum interference device **310**.

[0217] The configuration shown in FIG. **11** corresponds to a circuit structure that is an extension of the LHZ scheme. This circuit can embed a fully connected 6-bit logical model, using the LHZ scheme at most. This circuit can be considered an example of applying it to a problem represented by a fully connected 4-bit logical model, where several qubit devices **110** are used as readout qubit devices **112**.

[0218] Thus, a circuit capable of handling an LHZ model larger than the problem to be solved can be employed, and some of the qubit devices **110**, which are not utilized for solution search through quantum annealing, may be used as readout qubit devices **112**.

[0219] In the example shown in FIG. **11** also, the control unit **210** adiabatically changes the state until it reaches the coherent state, while simultaneously inputting a pump signal to the computational qubit device **111** to induce interactions. After the computational qubit device **111** reaches the coherent state, the control unit **210** inputs a pump signal to the readout qubit device **112**, inducing two-qubit interactions with the computational qubit device **111**. Then, the observation unit **220** reads out the bit value from the readout qubit device **112**.

[0220] The circuit structure of the information processing device **100** in the third example embodiment may be any circuit structure in which an LHZ model can be embedded, and is not limited to a specific network structure shape. For example, the information processing device **100** may be configured in the shape of a model based on the LHZ scheme, or may be configured in a square shape.

[0221] In the third example embodiment also, the information processing qubit is described as a computational qubit. However, the information processing qubit may also be used for purposes other than computation.

[0222] As described above, the information processing device **100** is configured with a circuit including a plurality of the qubit devices **110** on which an LHZ model, which is a model based on the LHZ scheme, is implemented, and four-qubit couplers **122** that are the couplers **120**, in a circuit having a larger number of bits in a logical model than a problem that is a target of quantum computing. The control unit **210** controls the information processing device **100** so that: one or more of the qubit devices **110** are operated as the computational qubit devices **111**; one or more of the qubit devices **110** other than the computational qubit devices **111** are operated as the readout qubit devices **112**; and one of the readout qubit devices **112** and one of the computational qubit devices **111** are two-qubit-coupled for each of the four-qubit couplers **122** coupling the readout qubit devices **112** with other qubit devices **110**.

[0223] According to the information processing device **100**, the readout qubit device **112** can be provided in the information processing device **100**, which implements a physical model obtained by converting a logical model using the LHZ scheme, without the necessity of separately adding a

qubit device **110** to be used as the readout qubit device **112**. Thus, in the information processing device **100**, it is expected that the bit value can be read out with relatively high accuracy from the LHZ scheme model. In this respect, it is expected that the information processing device **100** will be able to calculate with higher accuracy a solution to a problem represented by a fully connected logical model.

Fourth Example Embodiment

[0224] FIG. **12** is a diagram showing a configuration example of the information processing device according to at least one of the example embodiments. In the configuration shown in FIG. **12**, an information processing device **610** includes an information processing qubit device **611**, a readout qubit device **612**, and a coupler **613**.

[0225] With such a configuration, the coupler **613** couples the information processing qubit device **611** and the readout qubit device **612**. The readout qubit device **612** is coupled by the coupler **613** to the information processing qubit device **611** so as to assume the same bit value as the information processing qubit device **611**.

[0226] In the information processing device **610**, the control of the readout qubit device **612** does not need to be the same as that for an information processing qubit device **611**. In this respect, the information processing device **610** has a relatively higher degree of freedom in controlling the readout qubit device **612**. As a result, it is expected that the information processing device **610** will be able to control the readout qubit device **612** in a manner more suitable for reading out bit values, thereby making it possible to read out bit values with higher accuracy.

[0227] FIG. **13** is a diagram showing an example of a configuration of the information processing device **610**, where information processing qubit devices **611** and readout qubit devices **612** are denoted as qubit devices **614**, explicitly exemplifying that the information processing device **610** includes multiple qubit devices **614**.

[0228] The information processing device **610** includes, as qubit devices **614**, one or more information processing qubit devices and one or more readout qubit devices. The information processing qubit devices may be used as computational qubit devices.

Fifth Example Embodiment

[0229] FIG. **14** is a diagram showing a configuration example of the control device according to at least one of the example embodiments. In the configuration shown in FIG. **14**, a control device **620** includes a control unit **621**.

[0230] With such a configuration, the control unit **621** controls the information processing device including an information processing qubit device and a readout qubit device, and a coupler that couples the information processing qubit device and the readout qubit device, so that the readout qubit device is coupled by the coupler to the information processing qubit device so as to assume the same bit value as the information processing qubit device.

[0231] The control unit **621** serves as an example of the control means.

[0232] According to the control device **620**, the control of the readout qubit device does not need to be the same as that for an information processing qubit device, and the degree of freedom for controlling the readout qubit device is relatively high. As a result, it is expected that the control device **620** will be able to control the readout qubit device in a manner more suitable for reading out bit values, thereby making it possible to read out bit values with higher accuracy.

Sixth Example Embodiment

[0233] FIG. **15** is a diagram showing a configuration example of an information processing system according to at least one of example embodiments. In the configuration shown in FIG. **15**, an information processing system **630** includes an information processing device **631** and a control device **635**. The information processing device **631** includes an information processing qubit device **632**, a readout qubit device **633**, and a coupler **634**. The control device **635** includes a control unit **636**.

[0234] With such a configuration, the coupler **634** couples the information processing qubit device

632 and the readout qubit device **633**. The control unit **636** controls the information processing device **631** so that the readout qubit device **633** is coupled by the coupler **634** to the information processing qubit device **632** so as to assume the same bit value as the information processing qubit device **632**.

[0235] The control unit **636** serves as an example of the control means.

[0236] In the information processing system **630**, the control unit **636** does not need to control the readout qubit device **633** in the same manner as the information processing qubit device **632**. In this respect, the information processing system **630** has a relatively higher degree of freedom in controlling the readout qubit device **633**. As a result, in the information processing system **630**, it is expected to be able to control the readout qubit device **633** in a manner more suitable for reading out bit values, thereby making it possible to read out bit values with higher accuracy.

[0237] FIG. **16** is a diagram showing an example of a configuration of the information processing system **630**, where information processing qubit devices **632** and readout qubit devices **633** are denoted as qubit devices **637**, explicitly exemplifying that the information processing device **631** includes multiple qubit devices **637**.

[0238] The information processing device **631** includes, as qubit devices **637**, one or more information processing qubit devices and one or more readout qubit devices. The information processing qubit devices may be used as computational qubit devices.

Seventh Example Embodiment

[0239] FIG. **17** is a diagram showing an example of a processing procedure in a control method according to at least one of the example embodiments. The control method shown in FIG. **17** includes a step of controlling an information processing device (Step **S611**).

[0240] In the step of controlling an information processing device (Step **S611**), a control device that controls an information processing device including an information processing qubit device and a readout qubit device, and a coupler that couples the information processing qubit device and the readout qubit device, controls the information processing device, so that the readout qubit device is coupled by the coupler to the information processing qubit device so as to assume the same bit value as the information processing qubit device.

[0241] According to the control method shown in FIG. **17**, the control of the readout qubit device does not need to be the same as that for an information processing qubit device, and the degree of freedom for controlling the readout qubit device is relatively high. As a result, it is expected in the control method shown in FIG. **17** that it will be able to control the readout qubit device in a manner more suitable for reading out bit values, thereby making it possible to read out bit values with higher accuracy.

[0242] FIG. **18** is a diagram showing a configuration example of a computer according to at least one of the example embodiments. In the configuration shown in FIG. **18**, a computer **700** includes a CPU **710**, a primary storage device **720**, an auxiliary storage device **730**, an interface **740**, and a non-volatile recording medium **750**. Also, the computer **700** is connected to a quantum device **760** for communication. For example, as in the example of FIG. **4**, the computer **700** may be connected to the quantum device **760** for communication via a control unit that converts a control signal from the computer **700** into an analog control signal and transmits it to the quantum device **760**, and an observation unit that converts the analog signal from the quantum device **760** into a digital signal.

[0243] The quantum device **760** may be configured as part of the computer **700**. Alternatively, the quantum device **760** may be a component external to the computer **700**.

[0244] One or more of the control device **200**, the control device **620**, and the control device **635** mentioned above or part thereof may be implemented in the computer **700**. In such a case, operations of the respective processing units described above are stored in the auxiliary storage device **730** in the form of a program. The CPU **710** reads out the program from the auxiliary storage device **730**, loads it on the primary storage device **720**, and executes the processing described above according to the program. Moreover, the CPU **710** secures, according to the

program, memory storage regions corresponding to the respective storage units mentioned above, in the primary storage device **720**. Communication between each device and other devices is executed by the interface **740** having a communication function and communicating under the control of the CPU **710**. The interface **740** also has a port for the non-volatile recording medium **750**, and reads information from the non-volatile recording medium **750** and writes information to the non-volatile recording medium **750**.

[0245] The quantum device **760** is a device (circuit) that operates using a quantum state in quantum mechanics. The quantum device **760** operates as described above in each example embodiment for quantum computing, such as quantum annealing.

[0246] In the case where the control device **200** is implemented in the computer **700**, operations of the processing unit **235** and each component thereof are stored in the form of a program in the auxiliary storage device **730**. The CPU **710** reads out the programs from the auxiliary storage device **730**, loads them on the primary storage device **720**, and executes the processes described above, according to the programs.

[0247] Also, the CPU **710** secures a memory storage region in the primary storage device **720** for the storage unit **234**, according to the program. Communication with other devices performed by the communication unit **231** is executed by the interface **740** having a communication function and operating under the control of the CPU **710**. Display of images performed by the display unit **232** is executed by the interface **740** having a display device and displaying various images under the control of the CPU **710**. User operations are accepted through the operation input unit **233** by the interface **740** having an input device and accepting user operations under control of the CPU **710**.

[0248] In the case where the control device **620** is implemented in the computer **700**, operations of the control unit **621** are stored in the auxiliary memory storage device **730** in the form of a program. The CPU **710** reads out the programs from the auxiliary storage device **730**, loads them on the primary storage device **720**, and executes the processes described above, according to the programs.

[0249] Moreover, the CPU **710** secures a memory storage region in the primary storage device **720** for the processing to be performed by the control device **620**, according to the program.

Communication with other devices performed by the control device **620** is executed by the interface **740** having a communication function and operating under the control of the CPU **710**. Interaction between the control device **620** and the user is executed by the interface **740** having an input device and an output device, presenting information to the user through the output device under the control of the CPU **710**, and accepting user operations through the input device.

[0250] In the case where the control device **635** is implemented in the computer **700**, operations of the control unit **636** are stored in the auxiliary memory storage device **730** in the form of a program. The CPU **710** reads out the programs from the auxiliary storage device **730**, loads them on the primary storage device **720**, and executes the processes described above, according to the programs.

[0251] Moreover, the CPU **710** secures a memory storage region in the primary storage device **720** for the processing to be performed by the control device **635**, according to the program.

Communication with other devices performed by the control device **635** is executed by the interface **740** having a communication function and operating under the control of the CPU **710**. Interaction between the control device **635** and the user is executed by the interface **740** having an input device and an output device, presenting information to the user through the output device under the control of the CPU **710**, and accepting user operations through the input device.

[0252] Any one or more of the programs described above may be recorded in the non-volatile recording medium **750**. In such a case, the interface **740** may read the program from the non-volatile recording medium **750**. Then, the CPU **710** directly executes the program read by the interface **740**, or it may be temporarily stored in the primary storage device **720** or the auxiliary storage device **730** and then executed.

[0253] It should be noted that a program for executing some or all of the processes performed by the control device **200**, the control device **620**, and the control device **635** may be recorded on a computer-readable recording medium, and the program recorded on the recording medium may be read into and executed on a computer system, to thereby perform the processing of each unit. The “computer system” here includes an OS (operating system) and hardware such as peripheral devices.

[0254] Moreover, the “computer-readable recording medium” referred to here refers to a portable medium such as a flexible disk, a magnetic optical disk, a ROM (Read Only Memory), and a CD-ROM (Compact Disc Read Only Memory), or a storage device such as a hard disk built into a computer system. The above program may be a program for realizing a part of the functions described above, and may be a program capable of realizing the functions described above in combination with a program already recorded in a computer system.

[0255] The example embodiments have been described in detail. However, the specific configuration of the disclosure is not limited to the example embodiments, and may include designs and so forth that do not depart from the scope of the present disclosure. Moreover, the example embodiments described above may be combined with another example embodiment where appropriate. In the above example embodiment, the superconducting quantum interference device has been described with respect to an example in which the number of Josephson junctions is two. However, the number of Josephson junctions may be three or more.

[0256] According to the present disclosure, it is possible to enable a relatively higher degree of freedom of control over for a qubit device whose bit value is to be read out.

[0257] While preferred example embodiments of the disclosure have been described and illustrated above, it should be understood that these are exemplary of the disclosure and are not to be considered as limiting. Additions, omissions, substitutions, and other modifications can be made without departing from the scope of the present disclosure. Accordingly, the disclosure is not to be considered as being limited by the foregoing description, and is only limited by the scope of the appended claims.

[0258] A part or all of the example embodiments described above can be written as in the supplementary notes below, but is not limited thereto.

(Supplementary Note 1)

[0259] An information processing device comprising: [0260] an information processing qubit device and a readout qubit device, each configured with a Josephson parametric oscillator; and [0261] a coupler that couples the information processing qubit device and the readout qubit device. [0262] wherein the readout qubit device is coupled by the coupler to the information processing qubit device so as to assume the same bit value as the information processing qubit device.

(Supplementary Note 2)

[0263] The information processing device according to supplementary note 1, wherein the information processing qubit device is a computational qubit device.

(Supplementary Note 3)

[0264] The information processing device according to supplementary note 2, wherein of the readout qubit device and the computational qubit device coupled to each other, the readout qubit device has a nonlinearity lower than that of the computational qubit device.

(Supplementary Note 4)

[0265] The information processing device according to supplementary note 2 or 3, wherein the readout qubit device and the computational qubit device coupled to each other are excited at the same frequency.

(Supplementary Note 5)

[0266] The information processing device according to any one of supplementary notes 2 to 4, wherein a pump frequency of the computational qubit device is higher than twice a resonance frequency of the computational qubit device, and [0267] a pump frequency of the readout qubit

device is lower than twice a resonance frequency of the readout qubit device.

(Supplementary Note 6)

[0268] The information processing device according to supplementary note 2 or 3, wherein the readout qubit device and the computational qubit device coupled to each other are excited at the same pump frequency, [0269] a resonance frequency of the computational qubit device is lower than one-half the pump frequency, and [0270] a resonance frequency of the readout qubit device is higher than one-half the pump frequency.

(Supplementary Note 7)

[0271] The information processing device according to any one of supplementary notes 2 to 6, wherein the readout qubit device is excited after the computational qubit device reaches a coherent state.

(Supplementary Note 8)

[0272] The information processing device according to any one of supplementary notes 2 to 7, wherein the computational qubit device and one or more four-qubit couplers among the couplers form an LHZ scheme model.

(Supplementary Note 9)

[0273] A control device comprising [0274] a control means that controls an information processing device including an information processing qubit device and a readout qubit device, each configured with a Josephson parametric oscillator, and a coupler that couples the information processing qubit device and the readout qubit device, so that the readout qubit device is coupled by the coupler to the information processing qubit device so as to assume the same bit value as the information processing qubit device.

(Supplementary Note 10)

[0275] The control device according to supplementary note 9, wherein the information processing qubit device is a computational qubit device.

(Supplementary Note 11)

[0276] The control device according to supplementary note 10, wherein of the readout qubit device and the computational qubit device coupled to each other, the readout qubit device has a nonlinearity lower than that of the computational qubit device.

(Supplementary Note 12)

[0277] The control device according to supplementary note 10 or 11, wherein the control means excites the readout qubit device and the computational qubit device coupled to each other at the same frequency.

(Supplementary Note 13)

[0278] The control device according to any one of supplementary notes 10 to 12, wherein the control means controls the information processing device so that a pump frequency of the computational qubit device is of a value higher than twice a resonance frequency of the computational qubit device, and a pump frequency of the readout qubit device is of a value lower than twice a resonance frequency of the readout qubit device.

(Supplementary Note 14)

[0279] The control device according to supplementary note 10 or 11, wherein the control means controls the information processing device so that the readout qubit device and the computational qubit device coupled to each other are excited at the same pump frequency, a resonance frequency of the computational qubit device is lower than one-half the pump frequency, and a resonance frequency of the readout qubit device is higher than one-half the pump frequency.

(Supplementary Note 15)

[0280] The control device according to any one of supplementary notes 10 to 14, wherein the control means excites the readout qubit device after the computational qubit device reaches a coherent state.

(Supplementary Note 16)

[0281] The control device according to any one of supplementary notes 10 to 15, wherein the control means controls the information processing device in which the computational qubit device and one or more four-qubit couplers among the couplers form an LHZ scheme model.

(Supplementary Note 17)

[0282] The control device according to any one of supplementary notes 10 to 16, wherein the information processing device is configured with a circuit including a plurality of the qubit devices on which an LHZ model, which is a model based on the LHZ scheme, is implemented, and four-qubit couplers that are the couplers, in a circuit having a larger number of bits in a logical model than a problem that is a target of quantum computing, and [0283] the control means controls the information processing device so that one or more of the qubit devices are operated as the computational qubit devices, one or more of the qubit devices other than the computational qubit devices are operated as the readout qubit devices, and one of the readout qubit devices and one of the computational qubit devices are two-qubit-coupled for each of the four-qubit couplers coupling the readout qubit devices with other qubit devices.

(Supplementary Note 18)

[0284] An information processing system comprising: [0285] an information processing device; and [0286] a control device [0287] wherein the information processing device includes: [0288] an information processing qubit device and a readout qubit device, each configured with a Josephson parametric oscillator; and [0289] a coupler that couples the information processing qubit device and the readout qubit device, [0290] wherein the control device includes a control means that controls the information processing device so that the readout qubit device is coupled by the coupler to the information processing qubit device so as to assume the same bit value as the information processing qubit device.

(Supplementary Note 19)

[0291] The information processing system according to supplementary note 18, wherein the information processing qubit device is a computational qubit device.

(Supplementary Note 20)

[0292] The information processing system according to supplementary note 19, wherein of the readout qubit device and the computational qubit device coupled to each other, the readout qubit device has a nonlinearity lower than that of the computational qubit device.

(Supplementary Note 21)

[0293] The information processing system according to supplementary note 19 or 20, wherein the control means excites the readout qubit device and the computational qubit device coupled to each other at the same frequency.

(Supplementary Note 22)

[0294] The information processing system according to any one of supplementary notes 19 to 21, wherein the control means controls the information processing device so that a pump frequency of the computational qubit device is of a value higher than twice a resonance frequency of the computational qubit device, and a pump frequency of the readout qubit device is of a value lower than twice a resonance frequency of the readout qubit device.

(Supplementary Note 23)

[0295] The information processing system according to supplementary note 19 or 20, wherein the control means controls the information processing device so that the readout qubit device and the computational qubit device coupled to each other are excited at the same pump frequency, a resonance frequency of the computational qubit device is lower than one-half the pump frequency, and a resonance frequency of the readout qubit device is higher than one-half the pump frequency.

(Supplementary Note 24)

[0296] The information processing system according to any one of supplementary notes 19 to 23, wherein the control means excites the readout qubit device after the computational qubit device reaches a coherent state.

(Supplementary Note 25)

[0297] The information processing system according to any one of supplementary notes 19 to 24, wherein the control means controls the information processing device in which the computational qubit device and one or more four-qubit couplers among the couplers form an LHZ scheme model.

(Supplementary Note 26)

[0298] The information processing system according to any one of supplementary notes 19 to 25, wherein the information processing device is configured with a circuit including a plurality of the qubit devices on which an LHZ model, which is a model based on the LHZ scheme, is implemented, and four-qubit couplers that are the couplers, in a circuit having a larger number of bits in a logical model than a problem that is a target of quantum computing, and [0299] the control means controls the information processing device so that one or more of the qubit devices are operated as the computational qubit devices, one or more of the qubit devices other than the computational qubit devices are operated as the readout qubit devices, and one of the readout qubit devices and one of the computational qubit devices are two-qubit-coupled for each of the four-qubit couplers coupling the readout qubit devices with other qubit devices.

(Supplementary Note 27)

[0300] A control method comprising a step, performed by [0301] a control device that controls an information processing device comprising an information processing qubit device and a readout qubit device, each configured with a Josephson parametric oscillator, and a coupler that couples the information processing qubit device and the readout qubit device, of [0302] controlling the information processing device so that the readout qubit device is coupled by the coupler to the information processing qubit device so as to assume the same bit value as the information processing qubit device.

(Supplementary Note 28)

[0303] The control method according to supplementary note 27, wherein the information processing qubit device is a computational qubit device.

(Supplementary Note 29)

[0304] The control method according to supplementary note 28, wherein of the readout qubit device and the computational qubit device coupled to each other, the readout qubit device has a nonlinearity lower than that of the computational qubit device.

(Supplementary Note 30)

[0305] The control method according to supplementary note 28 or 29, wherein the step of controlling the information processing device includes a step of exciting the readout qubit device and the computational qubit device coupled to each other at the same frequency.

(Supplementary Note 31)

[0306] The control method according to any one of supplementary notes 28 to 30, wherein the step of controlling the information processing device includes a step of controlling the information processing device so that a pump frequency of the computational qubit device is of a value higher than twice a resonance frequency of the computational qubit device, and a pump frequency of the readout qubit device is of a value lower than twice a resonance frequency of the readout qubit device.

(Supplementary Note 32)

[0307] The control method according to supplementary note 28 or 29, wherein the step of controlling the information processing device includes a step of controlling the information processing device so that the readout qubit device and the computational qubit device coupled to each other are excited at the same pump frequency, a resonance frequency of the computational qubit device is lower than one-half the pump frequency, and a resonance frequency of the readout qubit device is higher than one-half the pump frequency.

(Supplementary Note 33)

[0308] The control method according to any one of supplementary notes 28 to 32, wherein the step

of controlling the information processing device includes a step of exciting the readout qubit device after the computational qubit device reaches a coherent state.

(Supplementary Note 34)

[0309] The control method according to any one of supplementary notes 28 to 33, wherein the step of controlling the information processing device includes a step of controlling the information processing device in which the computational qubit device and one or more four-qubit couplers among the couplers form an LHZ scheme model.

(Supplementary Note 35)

[0310] The control method according to any one of supplementary notes 28 to 34, wherein the information processing device is configured with a circuit including a plurality of the qubit devices on which an LHZ model, which is a model based on the LHZ scheme, is implemented, and four-qubit couplers that are the couplers, in a circuit having a larger number of bits in a logical model than a problem that is a target of quantum computing, and [0311] the step of controlling the information processing device includes a step of controlling the information processing device so that one or more of the qubit devices are operated as the computational qubit devices, one or more of the qubit devices other than the computational qubit devices are operated as the readout qubit devices, and one of the readout qubit devices and one of the computational qubit devices are two-qubit-coupled for each of the four-qubit couplers coupling the readout qubit devices with other qubit devices.

(Supplementary Note 36)

[0312] A program that causes [0313] a computer that controls an information processing device including an information processing qubit device and a readout qubit device, each configured with a Josephson parametric oscillator, and a coupler that couples the information processing qubit device and the readout qubit device, to execute a step of [0314] controlling the information processing device so that the readout qubit device is coupled by the coupler to the information processing qubit device so as to assume the same bit value as the information processing qubit device.

(Supplementary Note 37)

[0315] The program according to supplementary note 36, wherein the information processing qubit device is a computational qubit device.

(Supplementary Note 38)

[0316] The program according to supplementary note 37, wherein of the readout qubit device and the computational qubit device coupled to each other, the readout qubit device has a nonlinearity lower than that of the computational qubit device.

(Supplementary Note 39)

[0317] The program according to supplementary note 37 or 38, wherein the step of controlling the information processing device includes a step of causing the computer to excite the readout qubit device and the computational qubit device coupled to each other at the same frequency.

(Supplementary Note 40)

[0318] The program according to any one of supplementary notes 37 to 39, wherein the step of controlling the information processing device includes a step of causing the computer to execute control of the information processing device so that a pump frequency of the computational qubit device is of a value higher than twice a resonance frequency of the computational qubit device, and a pump frequency of the readout qubit device is of a value lower than twice a resonance frequency of the readout qubit device.

(Supplementary Note 41)

[0319] The program according to supplementary note 37 or 38, wherein the step of controlling the information processing device includes a step of causing the computer to execute control of the information processing device so that the readout qubit device and the computational qubit device coupled to each other are excited at the same pump frequency, a resonance frequency of the

computational qubit device is lower than one-half the pump frequency, and a resonance frequency of the readout qubit device is higher than one-half the pump frequency.
(Supplementary Note 42)

[0320] The program according to any one of supplementary notes 37 to 41, wherein the step of controlling the information processing device includes a step of causing the computer to execute excitation of the readout qubit device after the computational qubit device reaches a coherent state.
(Supplementary Note 43)

[0321] The program according to any one of supplementary notes 37 to 42, wherein the step of controlling the information processing device includes a step of causing the computer to execute control of the information processing device in which the computational qubit device and one or more four-qubit couplers among the couplers form an LHZ scheme model.
(Supplementary Note 44)

[0322] The program according to any one of supplementary notes 37 to 43, wherein the information processing device is configured with a circuit including a plurality of the qubit devices on which an LHZ model, which is a model based on the LHZ scheme, is implemented, and four-qubit couplers that are the couplers, in a circuit having a larger number of bits in a logical model than a problem that is a target of quantum computing, and [0323] the step of controlling the information processing device includes a step of causing the computer to execute control of the information processing device so that one or more of the qubit devices are operated as the computational qubit devices, one or more of the qubit devices other than the computational qubit devices are operated as the readout qubit devices, and one of the readout qubit devices and one of the computational qubit devices are two-qubit-coupled for each of the four-qubit couplers coupling the readout qubit devices with other qubit devices.

Claims

1. An information processing device comprising: a first Josephson parametric oscillator used as an information processing qubit element; a second Josephson parametric oscillator used as a readout qubit element, each configured with a Josephson parametric oscillator; and a coupler coupling the information processing qubit element and the readout qubit element, wherein a bit value indicated by the readout qubit element is same as a bit value indicated by the information processing qubit element.
2. The information processing device according to claim 1, wherein the information processing qubit element is a computational qubit element.
3. The information processing device according to claim 2, wherein of the readout qubit element and the computational qubit element coupled to each other, a nonlinearity of the readout qubit element is lower than a nonlinearity of the computational qubit element.
4. The information processing device according to claim 2, wherein the readout qubit element and the computational qubit element coupled to each other are excited at a same frequency.
5. The information processing device according to claim 2, wherein a pump frequency of the computational qubit element is higher than twice a resonance frequency of the computational qubit element, and wherein a pump frequency of the readout qubit element is less than twice a resonance frequency of the readout qubit element.
6. The information processing device according to claim 2, wherein the readout qubit element and the computational qubit element coupled to each other are excited at a same pump frequency, wherein the pump frequency is higher than twice a resonance frequency of the computational qubit element, and wherein the pump frequency is less than twice a resonance frequency of the readout qubit element.
7. The information processing device according to claim 2, wherein the readout qubit element is excited after the computational qubit element reaches a coherent state.

- 8.** The information processing device according to claim 2, wherein the computational qubit element and one or more four-qubit couplers among the couplers form a Lechner-Hauke-Zoller (LHZ) scheme model.
- 9.** A control device comprising a controller configured to control an information processing element including a first Josephson parametric oscillator used as an information processing qubit element, a second Josephson parametric oscillator used as a readout qubit element, and a coupler coupling the information processing qubit element and the readout qubit element, wherein the controller controls the readout qubit element and the information processing qubit element so that a bit value indicated by the readout qubit device is same as a bit value indicated by the information processing qubit element.
- 10.** A control method executed by a control device for controlling an information processing device including a first Josephson parametric oscillator used as an information processing qubit element, a second Josephson parametric oscillator used as a readout qubit element, and a coupler coupling the information processing qubit element and the readout qubit element, the method comprising controlling the readout qubit element and the information processing element so that a bit value indicated by the readout qubit element is same as a bit value indicated by the information processing qubit element.
- 11.** The control method according to claim 10, wherein the information processing qubit element is a computational qubit element.
- 12.** The control method according to claim 11, wherein of the readout qubit element and the computational qubit element coupled to each other, a nonlinearity of the readout qubit element is lower than a nonlinearity of the computational qubit element.
- 13.** The control method according to claim 11, further comprising exciting the readout qubit element and the computational qubit element coupled to each other at a same frequency.
- 14.** The control method according to claim 11, wherein a pump frequency of the computational qubit element is higher than twice a resonance frequency of the computational qubit element, and wherein a pump frequency of the readout qubit element is less than twice a resonance frequency of the readout qubit element.
- 15.** The control method according to claim 11, further comprising exciting the readout qubit element and the computational qubit element coupled to each other at a same pump frequency, wherein the pump frequency is higher than twice a resonance frequency of the computational qubit element, and wherein the pump frequency is less than twice a resonance frequency of the readout qubit element.
- 16.** The control method according to claim 11, further comprising exciting the readout qubit element after the computational qubit element reaches a coherent state.
- 17.** The control method according to claim 11, wherein the computational qubit element and one or more four-qubit couplers among the couplers form a Lechner-Hauke-Zoller (LHZ) scheme model.
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